

Samarth Sharma
24BAI10894

CRUD OPERATIONS:(MYSQL)

```
package    CRUD;    import    java.sql.Connection;    import
java.sql.PreparedStatement; import java.sql.ResultSet; public class
CRUD_Test    {    public    void    create_data(String    sl_no,String
name,String email){

        DB_Connection obj_DB_Connection=new DB_Connection();
        Connection connection=obj_DB_Connection.get_connection();

        PreparedStatement ps=null;
        try {

            String query="insert into student(sl_no,name,email) values
            (?,?)";; ps=connection.prepareStatement(query);

            ps.setString(1, sl_no); ps.setString(2, name); ps.setString(3,
            email); System.out.println(ps); ps.executeUpdate();

        } catch (Exception e) {

            System.out.println(e);

        }
    }

    public void read_data(String sl_no) {

        DB_Connection obj_DB_Connection = new DB_Connection();
        Connection connection = obj_DB_Connection.get_connection();
        PreparedStatement ps = null;
        ResultSet rs =
null;

        try {

            String query = "SELECT * FROM student WHERE sl_no =
            ?"; ps = connection.prepareStatement(query);

            ps.setString(1, sl_no);
```

```
        System.out.println(ps);
rs = ps.executeQuery();

if (rs.next()) {
    System.out.println("Sl no - " + rs.getString("sl_no"));
    System.out.println("Name - " + rs.getString("name"));
    System.out.println("Email - " + rs.getString("email"));
} else {
    System.out.println("No record found for sl_no: " + sl_no);
}

} catch (Exception e) {
    System.out.println(e);
}
}

// public void update_data(String sl_no,String new_sl_no,String name,String
email){
    DB_Connection obj_DB_Connection=new DB_Connection();
    Connection connection=obj_DB_Connection.get_connection();
    PreparedStatement ps=null;
    try {
        String query="update student set sl_no=?,name=?,email=? where
sl_no=?";

        ps=connection.prepareStatement(quer
y); ps.setString(1, new_sl_no); ps.setString(2,
name); ps.setString(3, email); ps.setString(4,
sl_no); System.out.println(ps);
ps.executeUpdate(); } catch (Exception e) {
    System.out.println(e);
```

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```
        }  
    }  
  
    // public void delete_data(String sl_no){  
        DB_Connection obj_DB_Connection=new DB_Connection();  
        Connection connection=obj_DB_Connection.get_connection();  
        PreparedStatement ps=null;  
        try {  
            String query="delete from student where sl_no=?";  
            ps=connection.prepareStatement(query);  
            ps.setString(1, sl_no);  
            System.out.println(p  
s); ps.executeUpdate(); }  
        catch (Exception e) {  
            System.out.println(e);  
        }  
    }  
  
    public void alter_table() {  
        DB_Connection obj_DB_Connection = new DB_Connection();  
        Connection connection = obj_DB_Connection.get_connection();  
        PreparedStatement ps = null;  
        try {  
            String query = "alter table student ADD  
City      VARCHAR(20)";      ps      =  
            connection.prepareStatement(query);  
  
            System.out.println(ps);  
  
            ps.executeUpdate();  
  
            System.out.println("Column added  
successfully.");
```

```
    } catch (Exception e) {  
        System.out.println(e);  
    }  
}  
  
public void truncate_data() {  
    DB_Connection obj_DB_Connection = new DB_Connection();  
    Connection connection = obj_DB_Connection.get_connection();  
    PreparedStatement ps = null;  
  
    try {  
        String query = "truncate table student";  
        ps = connection.prepareStatement(query);  
  
        System.out.println(ps);  
  
        ps.executeUpdate();  
        System.out.println("All records deleted successfully.");  
    } catch (Exception e) {  
        System.out.println(e);  
    }  
}  
  
public static void main(String[] args) {  
    CRUD_Test objTest=new CRUD_Test();  
    //insert    data    objTest.create_data("1",    "jinu",  
        "jinu@gmail.com");    objTest.create_data("2",  
        "jawad", "jawad@gmail.com");
```

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```
// read data

objTest.read_data("2");

// update data objTest.update_data("2", "22", "jawad2",
    "jawad2@gmail");

// delete data

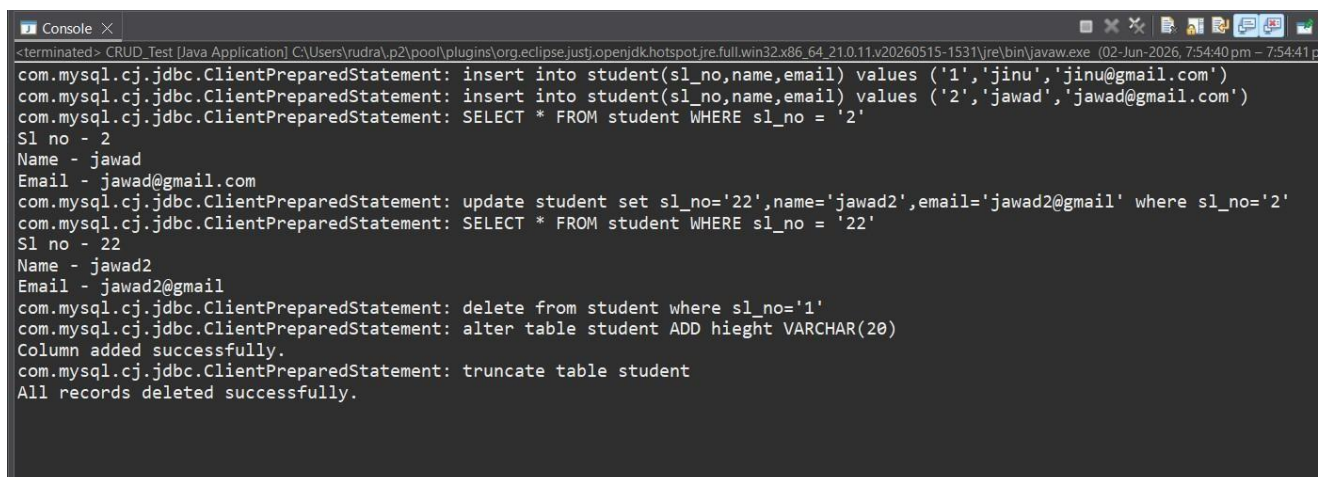
objTest.delete_data("1");

objTest.alter_table();

objTest.truncate_data();

    }

}
```

A screenshot of a Java console window titled 'Console X'. It displays the output of a Java application that interacts with a MySQL database. The output shows several SQL statements being executed: two INSERT statements for 'jtinu' and 'jawad', a SELECT statement for 's1_no = 2', an UPDATE statement for 's1_no = 22', another SELECT statement for 's1_no = 22', a DELETE statement for 's1_no = 1', an ALTER TABLE statement to add a 'hieght' column, and a TRUNCATE statement. The console also shows the resulting data for the first two rows: 'S1 no - 2', 'Name - jtinu', 'Email - jtinu@gmail.com' and 'S1 no - 22', 'Name - jawad2', 'Email - jawad2@gmail.com'. The final message is 'All records deleted successfully.'

```
<terminated> CRUD_Test [Java Application] C:\Users\rudra\p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.11.v20260515-1531\jre\bin\javaw.exe (02-Jun-2026, 7:54:40 pm - 7:54:41 p
com.mysql.cj.jdbc.ClientPreparedStatement: insert into student(s1_no,name,email) values ('1','jtinu','jtinu@gmail.com')
com.mysql.cj.jdbc.ClientPreparedStatement: insert into student(s1_no,name,email) values ('2','jawad','jawad@gmail.com')
com.mysql.cj.jdbc.ClientPreparedStatement: SELECT * FROM student WHERE s1_no = '2'
S1 no - 2
Name - jtinu
Email - jtinu@gmail.com
com.mysql.cj.jdbc.ClientPreparedStatement: update student set s1_no='22',name='jawad2',email='jawad2@gmail' where s1_no='2'
com.mysql.cj.jdbc.ClientPreparedStatement: SELECT * FROM student WHERE s1_no = '22'
S1 no - 22
Name - jawad2
Email - jawad2@gmail.com
com.mysql.cj.jdbc.ClientPreparedStatement: delete from student where s1_no='1'
com.mysql.cj.jdbc.ClientPreparedStatement: alter table student ADD hieght VARCHAR(20)
Column added successfully.
com.mysql.cj.jdbc.ClientPreparedStatement: truncate table student
All records deleted successfully.
```

MongoDB Connection: package mongodb2;

import com.mongodb.MongoClient; import

com.mongodb.client.MongoDatabase;

public class Test1 { public static void

main(String[] args) {

 MongoClient mongoClient = new

 MongoClient("localhost", 27017);

 MongoDatabase database = mongoClient.getDatabase("student_db");

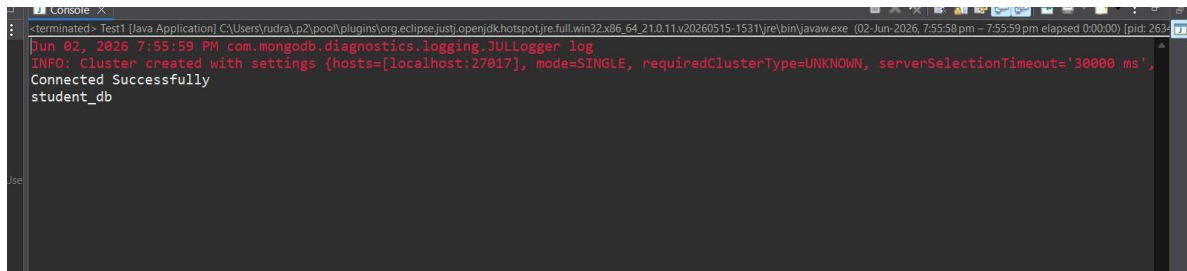
 System.out.println("Connected

 Successfully"); System.out.println(database.getName());

 }

}

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```
<terminated> Test1 [Java Application] C:\Users\rudra\p2\pool\plugins\org.eclipse.justi.openjdk hotspot.jre.full.win32.x86_64.21.0.11.v20260515-1531\jre\bin\javaw.exe (02-Jun-2026, 7:55:58 pm - 7:55:59 pm elapsed 0:00:00) [pid: 263]
Jun 02, 2026 7:55:59 PM com.mongodb.diagnostics.logging.JULLogger log
INFO: Cluster created with settings {hosts=[localhost:27017], mode=SINGLE, requiredClusterType=UNKNOWN, serverSelectionTimeout='30000 ms'},
Connected Successfully
student_db
```